

Seeing What It Looks Like When Interacting with XR-NEDs

Yuwei Wu

School of Mechatronic Engineering
and Automation
Shanghai University
Shanghai, China
Shangda General Intelligent Robotics
Research Institute
Shanghai, China
23721759@shu.edu.cn

Xinlei Li

School of Mechatronic Engineering
and Automation
Shanghai University
Shanghai, China
22120660666@shu.edu.cn

Xi Ye

School of Mechatronic Engineering
and Automation
Shanghai University
Shanghai, China
qianqianxiaokeai@shu.edu.cn

Xinyue Hu

School of Mechatronic Engineering
and Automation
Shanghai University
Shanghai, China
3157096737@shu.edu.cn

Yunfei Liu

School of Mechatronic Engineering
and Automation
Shanghai University
Shanghai, China
a5068865@shu.edu.cn

Yifan Peng

the University of Hong Kong
Hong Kong, Hong Kong
evanpeng@hku.hk

Xinxing Xia*

School of Mechatronic Engineering
and Automation
Shanghai University
Shanghai, China
Shangda General Intelligent Robotics
Research Institute
Shanghai, China
lygxia@gmail.com

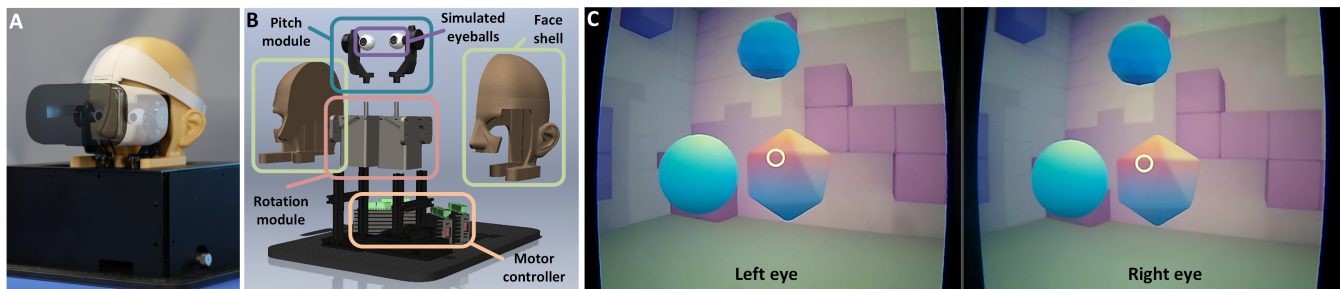


Figure 1: BinocuSim is a novel human eye simulator, mimicking human eye binocular motion and perception when wearing an extended reality (XR) near-eye display (A). Our simulator consists of a motor controller, a rotation module, a pitch module, a face shell, and two “eyeballs” with built-in cameras (B). The user can observe the XR scene perceived by each of BinocuSim’s two eyeballs through a graphical user interface (C). BinocuSim revolutionizes human-computer interaction (HCI) by merging XR near-eye displays with robotics, creating the first truly immersive human-robot cognitive partnership.

*Corresponding author

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Abstract

Comprehensively simulating and characterizing the perceptual behavior with extended reality near-eye displays (XR-NEDs) is essential yet often slow, costly, and largely dependent on human subjects. We present *BinocuSim*, a novel human vision simulator that replicates natural human eye movements while capturing and displaying the wearer’s XR-NED view via a customized GUI and a mounted XR-NED system. With achieving a mean gaze point

error of just 0.79 mm, *BinocuSim* can deliver high-fidelity, human-like perception and enable scalable evaluation of XR-NEDs across diverse human-computer interaction scenarios.

Keywords

Extended reality, near-eye displays, human perception simulation, human-computer interaction

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1 Introduction

XR-NEDs are transforming a wide range of disciplines [Koulieris et al. 2019]. Yet, due to the unique characteristics of these devices and individual variations in human vision, we lack precise knowledge of what scene information the wearer perceives at any given moment. This makes it difficult to reproduce the visual details within their field of view, turning the user's experience into a perceptual "black box".

To address these challenges, we present *BinocuSim*, a novel human vision simulator that integrates XR-NEDs with a robotic eye movement device to accurately reproduce what the wearer sees (Figure 1(A)). We design and build dedicated hardware and motion control algorithms to replicate human eye behaviors—including saccades, smooth pursuit, fixations, and micro-saccades—with high fidelity. Moreover, a custom GUI renders the captured XR scene in real time, enabling external observers to experience the user's perspective. *BinocuSim* not only achieves unprecedented perceptual realism but also introduces a new paradigm for visual sharing and human-computer interaction in XR, and further guides the generation and production of XR-NED's display content, making it more compatible with the human eye viewing characteristics.

2 Design and Implementation

To ensure *BinocuSim* remains affordable and modular, most structural components are either 3D-printed or sourced from off-the-shelf, avoiding proprietary hardware. As shown in Figure 1(B), each simulated eyeball module contains a miniature camera and is mounted on two compact actuators via custom connectors. This specific design provides two degrees of freedom per eye— $\pm 20^\circ$ horizontally (adduction/abduction) and $\pm 15^\circ$ vertically (elevation/depression)—covering the majority of natural eye movements. Compared to traditional cable-driven [Alitappeh et al. 2024; Rajendran et al. 2023] or multi-link bionic simulators [Cheng et al. 2019; Liu et al. 2015], *BinocuSim* achieves comparable binocular kinematics with fewer components and a DIY-friendly build. See Supplementary Material for detailed hardware specifications.

Each eye's motor controller is driven by a microcontroller running our custom motion control firmware. Gaze data from a commercial eye tracker (Pupil Core [Kassner et al. 2014]) is streamed via USB and parsed in real time. Users can either record their own binocular gaze trajectories for later play back or control *BinocuSim* live using real-time tracker input. The control loop accurately replicates

smooth pursuit, saccades, and fixations, including micro-saccades (see Supplementary Material for details).

To close the loop on "*what the eyes see*," we developed a lightweight GUI (Figure 1(C)) that displays the live feed from each simulated eyeball, overlaid with the corresponding XR content. As *BinocuSim* replays recorded eye movements or follows live tracker input, observers can directly see how the XR-NED scene maps onto the user's visual field (Figure 2). This setup enables rigorous, human-simulated validation of display and rendering strategies, while offering a novel solution to the XR "black-box" problem—making the user's visual experience observable and analyzable.



Figure 2: By combining the XR-NED with *BinocuSim*, the observer has direct access to the user's perceptual environment, which preserves the real operational experience for the user and resolves the black box where the observer cannot really "see" what the wearer sees.

3 Experimental results

To evaluate *BinocuSim*'s accuracy, we designed a test scenario using the cookie-theft image [Goodglass and Kaplan 1983]. Here, we invited 8 participants to help us evaluate the performance of *BinocuSim*, including 4 female and 4 male. To capture the diverse eye movements of participants, we designed our scenario and test images based on the visual stimuli conditions of different types of natural eye movements, so that the elements in the images could stimulate participants to produce eye movements corresponding to saccades, fixations, and smooth pursuits. During simulated binocular eye movements, *BinocuSim* achieved a mean gaze point error of 0.79 mm compared to the ground truth (Figure 3). Additional experimental results are provided in the Supplementary Material. These findings confirm that *BinocuSim* accurately replicates human eye movements during XR-NED interaction while providing real-time visualization of human-like perception.

Conclusion and Discussion. *BinocuSim* is a scalable human vision simulator designed for XR-NED interaction, built on an affordable mechanical platform and tightly integrated motion control algorithms that intuitively mimic biologically plausible eye movements. A custom GUI allows users to control the system via an eye tracker while visualizing the perceived environment, introducing a new paradigm for HCI in XR and robotics. With a mean gaze point error below 0.8 mm, *BinocuSim* meets the precision demands of XR-NED applications. Future works will focus on testing dynamic

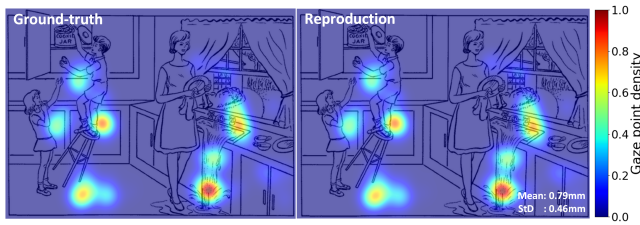


Figure 3: Heatmap shows the results of BinocuSim’s eye movement reproduction based on human-driven gaze inputs. In our test scenario with cookie-theft image[Goodglass and Kaplan 1983], BinocuSim achieved a gaze points mean error of 0.79 mm with a standard deviation of 0.46 mm when simulating human binocular eye movements and perception.

XR scenarios and variations in brightness, with the aim of enhancing BinocuSim’s alignment with complex XR application contexts and further improving its robustness.

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